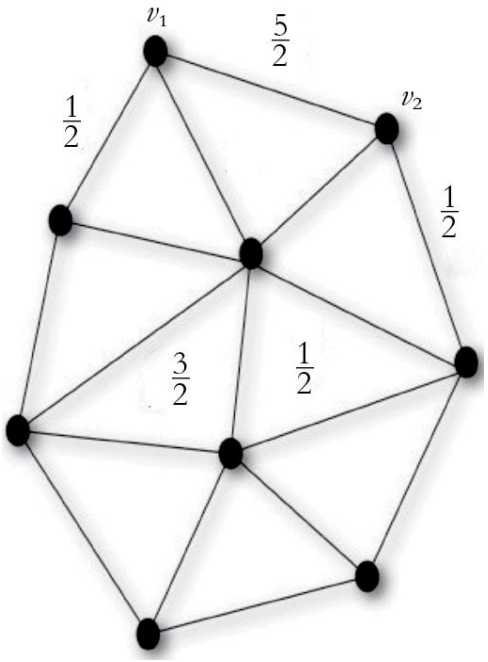


Figure 6.4 A spin network

The quanta of gravity, that is, are not *in* space, *they are themselves space*. The spin networks which describe the quantum structure of the gravitational field are not immersed in space; they do not inhabit a space. The location of single quanta of space is not defined with regard to something else but only by the links and the relation these express.

If I step from grain to grain along the links until I complete a circuit and return to the grain from which I started, I will have made a 'loop'. These are the original loops of the loop theory. In Chapter 4 I showed that the curvature of space may be measured by looking at whether an arrow transported across a closed circuit returns pointing in the same direction, or turned. The mathematics of the theory determines this curvature for every closed circuit on a spin network, and this makes it possible to evaluate the curvature of spacetime, and hence the force of the gravitational field, from the structure of a spin network.[fn38]

Now, quantum mechanics is more than granularity. There is also the fact that evolution is probabilistic – the way in which the spin networks evolve is random. I'll speak about this in the next chapter, devoted to time.

And there is the fact that what matters is not how things *are*, but rather how they *interact*. Spin networks are not entities; they describe the effect of space upon things. Just as an electron is in no place but diffused in a cloud of probability in all places, space is not actually formed by a single specific spin network but rather by a cloud of probabilities over the whole range of all possible spin networks.